LIBRARY BORROWING SYSTEM

System Analysis and Design — Mini-Project Final Report

Capstone 1: System Analysis and Design | Instructor: Prof. S. S. Kim

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This report presents the five required deliverable areas. Diagram spaces and file references are intentionally left for the team to complete before submission. The report documents the proposed system design; it does not claim that software has been implemented.

1. Project Plan

1.1 Project background and objectives

Small and medium-sized libraries may use paper records or disconnected tools to manage books, members, loans and returns. This can cause incorrect availability information, slow transactions and difficulty identifying overdue loans. The proposed Library Borrowing System centralizes these records and supports library staff in routine circulation activities.

The project objectives are to document system requirements, model borrowing and return processes, describe system functions, and develop conceptual and logical data models. The work covers the planning, analysis and design stages of the systems development life cycle.

1.2 Scope and users

In scope	Users
Member registration; catalog search and availability; loans and returns; due dates and overdue monitoring; books and physical copies; borrowing history and reports; staff authentication.	Library Member / Student; Librarian / Library Staff; Library Manager; System Administrator.

Out of scope: reservations, fines and payments, renewals, e-books, inter-library loans, purchasing and automatic e-mail/SMS notifications. Members may search the catalog without logging in; staff handle loans and returns at the desk.

1.3 Business rules

ID	Rule
BR1	Loan period: 14 days.
BR2	Maximum of five active loans per member.
BR3	Only Active members may borrow.
BR4	Only Available copies may be borrowed; a copy may have only one open loan.
BR5	Loans become overdue after the due date; the status is checked daily.
BR6	Members with overdue loans cannot borrow until the overdue loan is returned.
BR7	Damaged returned copies remain unavailable.

These are design assumptions adopted for this course project, not rules verified with an operating library.

Project repository: GitHub repository

2. Business Process Models (BPM)

Three BPMN process models describe the order of activities and the responsibilities of members, librarians, the library system and the manager. They use events, tasks, gateways and lanes. The previously drafted BPM sequences are retained without modification.

2.1 BPM1 — Borrow Book

Lanes: Member, Librarian, Library System. A member requests a loan and provides the card and copy. The librarian scans the member card; the system checks eligibility. If eligible, the copy is scanned and availability checked. If available, the system creates a loan with a 14-day due date and marks the copy On Loan. The librarian hands over the book and states the due date. Ineligible members and unavailable copies follow separate refusal paths.

BPMN diagrams (BPM1·BPM3): Editable Draw.io file | PNG export

2.2 BPM2 — Return Book

Lanes: Member, Librarian, Library System. The member returns the copy; the librarian scans it and the system locates the active loan. If none exists, the return is rejected for record checking. Otherwise, the librarian inspects the copy, the system closes the loan with a return date, and the copy is marked Available or Damaged. The librarian confirms the return.

BPM2 is included in the consolidated BPMN files linked above.

2.3 BPM3 — Monitor Overdue Loans

Lanes: Library System, Librarian, Library Manager. A daily timer starts the check. The system finds loans past their due dates. If none exist, the process ends. Otherwise, overdue statuses are updated and a list and summary are produced. The librarian reviews the list and contacts members by phone or in person, while the manager reviews the summary. Both branches complete before the process ends.

BPM3 is included in the consolidated BPMN files linked above.

These BPMs address the original manual-record problems: eligibility and availability checks become consistent, returns update records immediately, and overdue loans are identified systematically.

3. UML Use Case Modeling

One overall UML Use Case Diagram shows the functions offered by the system and the actors who interact with them. A use case diagram is a functional view, not a chronological workflow.

3.1 Actors

Actor	Role
Library Member	Searches catalog and participates in borrowing and returning at the desk.
Librarian	Registers members; handles circulation, books and copies; reviews overdue records.
Library Manager	Reviews overdue loans, history and reports.
System Administrator	Manages staff accounts and access.

3.2 Use case inventory

ID	Use case	Purpose
UC1	Borrow Book	Create a loan after eligibility and availability checks.
UC2	Return Book	Close the loan and update the copy status.
UC3	Register Member	Create a member record.
UC4	Manage Books and Copies	Maintain catalog and physical inventory.
UC5	Track Overdue Loans	Identify overdue loans and review results.
UC6	Generate Reports and Borrowing History	Show circulation history and summary reports.
UC7	Search Catalog and Check Availability	Find titles and available copies.
UC8	Log In	Authenticate staff and administrators.
UC9	Manage User Accounts	Maintain staff accounts.
UC10	Verify Member Eligibility	Check membership, limits and overdue status.
UC11	Verify Password	Validate login credentials.
UC12	Report Damaged Copy	Handle damage discovered during return.

3.3 Diagram structure and relationships

Draw one system-boundary rectangle labeled Library Borrowing System. Place the four actors outside and the twelve use cases inside. Connect Member to UC1, UC2 and UC7; Librarian to UC1–UC6 and UC8; Manager to UC5, UC6 and UC8; Administrator to UC8 and UC9. Member participation in UC1 and UC2 is at the service desk; the librarian operates the transaction.

Use dashed UML relationships: UC1 «include» UC10; UC8 «include» UC11; UC12 «extend» UC2. The first two are required supporting actions; the last occurs only when a returned copy is damaged. UC5 is a daily automated process whose results are reviewed by staff and the manager.

Overall UML Use Case Diagram: Editable Draw.io file | PNG export

4. Chen Entity Relationship Diagrams

Four conceptual Chen ERDs describe the main entities and their relationships. Use rectangles for entities, ovals for attributes, diamonds for relationships, underlining for identifiers and 1/M cardinalities. The four diagrams are views of one information model.

4.1 Chen ERD 1 — Book Catalog

Entities: BOOK (BookID, ISBN, Title, PublicationYear); AUTHOR (AuthorID, AuthorName); CATEGORY (CategoryID, CategoryName). Relationships: BOOK — Written By — AUTHOR (M:N); CATEGORY — Classifies — BOOK (1:M). This represents book titles, authors and categories without mixing them with individual physical copies.

Chen ERDs (all four diagrams): Editable Draw.io file | PNG export

4.2 Chen ERD 2 — Book Inventory

Entities: BOOK (BookID, Title); BOOK_COPY (CopyID, Barcode, CopyStatus); SHELF_LOCATION (LocationID, ShelfCode, Section). Relationships: BOOK — Has — BOOK_COPY (1:M); SHELF_LOCATION — Stores — BOOK_COPY (1:M). Individual copies have their own barcode and availability status.

Chen ERDs 2·4 are included in the consolidated files linked above.

4.3 Chen ERD 3 — Members and Staff

Entities: MEMBER (MemberID, Name, Email, PhoneNumber, MemberStatus); STAFF_ACCOUNT (StaffID, Username, PasswordHash, StaffName, AccountStatus); ROLE (RoleID, RoleName). Relationships: STAFF_ACCOUNT — Registers — MEMBER (1:M); ROLE — Assigned To — STAFF_ACCOUNT (1:M). PhoneNumber may be drawn as a multivalued (double-oval) attribute.

Chen ERDs 2·4 are included in the consolidated files linked above.

4.4 Chen ERD 4 — Book Circulation

Entities: MEMBER (MemberID, Name); BOOK_COPY (CopyID, CopyStatus); LOAN (LoanID, BorrowDate, DueDate, ReturnDate, LoanStatus, DaysOverdue); STAFF_ACCOUNT (StaffID, StaffName). MEMBER, BOOK_COPY and STAFF_ACCOUNT each relate 1:M to LOAN over time. DaysOverdue is derived and may be shown as a dashed oval. Each copy can have many historical loans but only one open loan.

Chen ERDs 2·4 are included in the consolidated files linked above.

5. IE Entity Relationship Diagrams

The four IE ERDs express the same data as relational tables using primary keys (PK), foreign keys (FK) and Crow’s Foot cardinalities. Junction tables and separate tables for multivalued attributes are introduced where necessary.

5.1 IE ERD 1 — Book Catalog

Table	Key fields and other attributes
BOOK	BookID (PK), ISBN, Title, PublicationYear, CategoryID (FK)
AUTHOR	AuthorID (PK), AuthorName
CATEGORY	CategoryID (PK), CategoryName
BOOK_AUTHOR	BookID (PK/FK), AuthorID (PK/FK)

CATEGORY 1:M BOOK; BOOK 1:M BOOK_AUTHOR; AUTHOR 1:M BOOK_AUTHOR. BOOK_AUTHOR resolves the conceptual many-to-many author relationship.

IE ERDs (all four diagrams): Editable Draw.io file | PNG export

5.2 IE ERD 2 — Book Inventory

Table	Key fields and other attributes
BOOK	BookID (PK), Title (shown here as reference to the shared BOOK table)
BOOK_COPY	CopyID (PK), BookID (FK), LocationID (FK), Barcode, CopyStatus
SHELF_LOCATION	LocationID (PK), ShelfCode, Section

BOOK 1:M BOOK_COPY and SHELF_LOCATION 1:M BOOK_COPY. Each copy references exactly one title and one shelf location.

IE ERDs 2·4 are included in the consolidated files linked on the previous page.

5.3 IE ERD 3 — Members and Staff

Table	Key fields and other attributes
MEMBER	MemberID (PK), Name, Email, MemberStatus, RegisteredByStaffID (FK)
MEMBER_PHONE	MemberID (PK/FK), PhoneNumber (PK)
STAFF_ACCOUNT	StaffID (PK), Username, PasswordHash, StaffName, AccountStatus, RoleID (FK)
ROLE	RoleID (PK), RoleName

ROLE 1:M STAFF_ACCOUNT; STAFF_ACCOUNT 1:M MEMBER; MEMBER 1:M MEMBER_PHONE. MEMBER_PHONE implements the multivalued phone-number attribute without repeated columns.

IE ERDs 2·4 are included in the consolidated files linked on the previous page.

5.4 IE ERD 4 — Book Circulation

Table	Key fields and other attributes
MEMBER	MemberID (PK), Name (shared entity)
BOOK_COPY	CopyID (PK), CopyStatus (shared entity)
STAFF_ACCOUNT	StaffID (PK), StaffName (shared entity)
LOAN	LoanID (PK), MemberID (FK), CopyID (FK), BorrowedByStaffID (FK), ReturnedByStaffID (FK, nullable), BorrowDate, DueDate, ReturnDate (nullable), LoanStatus

MEMBER 1:M LOAN and BOOK_COPY 1:M LOAN. STAFF_ACCOUNT connects to LOAN twice: issuing staff (required) and returning staff (optional until return). The derived DaysOverdue value is calculated from dates rather than stored as an independent column.

IE ERDs 2·4 are included in the consolidated files linked on the previous page.

6. Consistency, Methodology and Conclusion

6.1 Connections across deliverables

Process or function	Use cases	Data involved
Borrowing (BPM1)	UC1, UC7, UC10	MEMBER, BOOK_COPY, LOAN
Returns (BPM2)	UC2, UC12	BOOK_COPY, LOAN, STAFF_ACCOUNT
Overdue monitoring (BPM3)	UC5, UC6	LOAN, MEMBER
Member and catalog management	UC3, UC4, UC7	MEMBER, BOOK, BOOK_COPY, AUTHOR, CATEGORY
Access control	UC8, UC9, UC11	STAFF_ACCOUNT, ROLE

The same entities are referenced across several module diagrams and must not be duplicated in a combined database. A copy can have multiple past loans but no more than one open loan. Business rules such as loan limits and overdue restrictions require system checks; cardinalities alone do not enforce them.

6.2 Methods and source materials

The report follows the course's SDLC structure and its distinction between process (BPM), functional (UML use case), conceptual data (Chen ERD) and logical data (IE ERD) perspectives. UML actor/use-case notation, BPMN events and gateways, and IE Crow's Foot relationships are applied to communicate the design. Course materials used: 2nd lecture (SDLC); 3rd lecture (Requirements Analysis); 4th–5th lectures (Conceptual Modeling); 6th-week materials (Use Case, BPM, Chen ERD, IE ERD and SQL examples).

6.3 Conclusion

The proposed Library Borrowing System covers registration, catalog management, circulation, overdue tracking, reports and staff access. The project plan establishes its scope and rules; the three BPMs describe operational workflows; one UML diagram presents system functions; and four Chen plus four IE ERDs document the conceptual and logical data structures. These deliverables provide a consistent design foundation. Software development, testing and deployment are not included in this mini-project.

Final repository / submission folder: GitHub repository

Final report file name: Library_Borrowing_System_Final_Report_Hexagon_Completed.pdf